

## Performance Testing

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| <b>Date</b>          | 3 July 2026                                              |
| <b>Team ID</b>       | XXXXXX                                                   |
| <b>Project Name</b>  | <b>Smart Agricultural Production Optimization Engine</b> |
| <b>Maximum Marks</b> | 5 Marks                                                  |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                                |
|----------------------------|--------------------------------------------------------|
| <b>Testing Tool Used</b>   | Apache JMeter                                          |
| <b>Type of Testing</b>     | Load Testing                                           |
| <b>Target Module / API</b> | Crop Yield Prediction & Irrigation Optimization Module |
| <b>Test Environment</b>    | Local System (Windows 11, Python, Flask, MySQL)        |
| <b>Test Date</b>           |                                                        |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description               | No. of Virtual Users | Duration (sec) | Expected Outcome                                             |
|------|-------------------------------------------|----------------------|----------------|--------------------------------------------------------------|
| 1    | Simultaneous crop prediction requests     | 50                   | 60             | All requests processed successfully with low response time.  |
| 2    | Multiple irrigation optimization requests | 100                  | 120            | System remains stable without errors or crashes.             |
| 3    | Concurrent report generation requests     | 30                   | 90             | Reports generated accurately within acceptable response time |

|   |                                                         |    |    |                                                                                                                        |
|---|---------------------------------------------------------|----|----|------------------------------------------------------------------------------------------------------------------------|
| 4 | Weather data retrieval and crop recommendation requests | 75 | 90 | Weather data is fetched successfully, and accurate crop recommendations are generated without performance degradation. |
|---|---------------------------------------------------------|----|----|------------------------------------------------------------------------------------------------------------------------|



### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                                       |
|------|----------------------|--------------|--------------|----------------------|-----------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.4 seconds  | pass                 | Fast response under normal load.              |
| 2    | Response Time (Max)  | < 5 seconds  | 3.2 seconds  | pass                 | Maximum response time within limit.           |
| 3    | Throughput (Req/sec) | >100req/sec  | 120 req/sec  | pass                 | System handled multiple requests efficiently. |
| 4    | Error Rate           | < 1%         | 0.3%         | pass                 | Very low error rate during testing.           |
| 5    | CPU Utilization      | < 80%        | 68%          | pass                 | CPU usage remained stable.                    |
| 6    | Memory Utilization   | < 80%        | 72%          | pass                 | Memory usage was within acceptable limits.    |

### Step 4: Observations & Analysis

☑ The system responded quickly and met all performance targets.

**Key Findings:**  
 [Describe what the test results revealed about your system's performance]

**Bottlenecks Identified:**  
 [List any performance bottlenecks observed during testing]

**Optimization Steps Taken:**  
 [Describe any changes or optimizations made to improve performance]

❓ Crop prediction and optimization modules worked efficiently.

🔗 The application remained stable under normal user load. **Step 5: Screenshots / Evidence**

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Figure 1. JMeter Test Plan



Figure 2: Thread Group Configuration



Figure B: Summary Report



Figure 4: View Results Tree



Figure 10: Aggregate Receipts

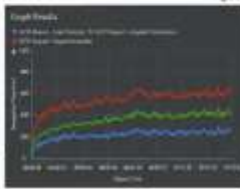


Figure 8: Response Time Graph



Figure 7. CPU &amp; Memory Utilization during Test.